MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set B | Duration: 3 Hours | Max Marks: 37 | Date: 11-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (6 Questions - 12 Marks)

Q1. [2 marks] (Medium)

Here are 10 unique exam questions for the Data Structures (CS101) Mid Term Examination, Set B:

Q2. [2 marks] (Easy)

In a singly linked list, what value is typically stored in the 'next' pointer of the very last node?

Options:
A) The address of the head node B) A special 'end' sentinel value C) NULL D) The address of the first node in memory

Q3. [2 marks] (Easy)

When implementing a stack using an array, which of the following conditions correctly indicates that the stack is full? (Assume MAX_SIZE is the array capacity and top points to the last inserted element's index.)

Options:
A) top == 0 B) top == MAX_SIZE - 1 C) top == -1 D) top < MAX_SIZE

Q4. [2 marks] (Easy)

Which operation is used to add an element to the rear of a queue?

Options:
A) Pop B) Push C) Enqueue D) Dequeue

Q6. [2 marks] (Medium)

Which data structure is primarily used to manage function calls and their local variables during program execution?

Options:
A) Queue B) Singly Linked List C) Stack D) Hash Table

Q8. [2 marks] (Hard)

Consider an empty stack. What will be the top element after the following sequence of operations: PUSH(5), PUSH(10), POP(), PUSH(15), PUSH(20), POP()?

Options:
A) 5 B) 10 C) 15 D) 20

Section 2: Short Answer (5 Questions - 25 Marks)

Q5. [5 marks] (Medium)

Briefly describe the steps required to insert a new node at the beginning of a singly linked list, given the new node's data and a pointer to the current head of the list.

Q7. [5 marks] (Medium)

State two distinct advantages of using a Doubly Linked List over a Singly Linked List.

Q9. [5 marks] (Medium)

Differentiate between a Stack and a Queue based on their fundamental data access principles and one common application for each.

Q10. [5 marks] (Hard)

Outline the algorithm to reverse a singly linked list iteratively. Assume you have a pointer to the head of the list.

Q11. [5 marks] (Hard)

In a linear array-based queue implementation, explain the condition under which the queue might appear "full" even if there are empty slots available, and how this issue is typically resolved without resizing the array.

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